CYBERMIND AI

What it was and what changed

What it was before

The first version was a threat detection and lockdown system. It watched user sessions on a demo bank app (VaultApp) and checked behaviour like typing speed, mouse movement, IP address and device. A set of rules plus an Isolation Forest model gave each session a risk level, and risky sessions were locked until an admin unlocked them. It had these parts:

- **Attack simulator:** buttons that fire one fake attack at a session, for example an IP change or an API request attack, to show the system reacting.
- **IP change detection:** if a session suddenly came from a different public IP, it was locked (later changed to ask for extra verification).
- **API burst detection:** too many requests in a short time triggered a lockdown.
- **RAG assistant (CyberMind Intel):** a chat assistant that answered security questions from uploaded documents using a vector database.
- **Admin console:** live alerts, logs, session list, lock and unlock.

What it does now

It still does all of that, but it now protects both moments of an account takeover: the login (before the attacker gets in) and the session (after someone steals it). Every decision is allow, ask for extra proof, contain in a sandbox, or block, and each one comes with the reasons in plain words so the admin can see why. A language model is never used to make or change a decision.

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What changed

Login attack detection (new): logins are grouped by the attacker's browser or script fingerprint instead of IP address, so an attack that uses thousands of different IPs is still caught as one campaign. It also uses decoy accounts that no real user would ever touch, and checks passwords against a known-breached list without ever storing the password.

Attack simulator got a Realistic Attack Lab: instead of one fake event, it replays whole attacks (rotating IPs, password spraying, slow attacks, session theft) mixed with normal users, and shows how many attacks were caught, how many got through, and how many normal users were wrongly bothered.

IP change detection is smarter: an IP or device change is now treated as one piece of evidence and compared with how that user normally behaves, not an automatic lock.

API burst: still there, but now it is one signal among others with a written reason.

Phishing email analysis (new): rules plus an optional AI second opinion, with a downloadable PDF report.

Feature 1, Phishing Poison Tracing: from an analysed phishing email, the analyst plants a fake username and password. If anyone ever tries it on our login, the attacker's whole campaign is blocked and a timeline shows email analysed, credential planted, credential used, campaign blocked.

Feature 2, Shadow Mode: when we are not sure a session is stolen, we do not lock it. The person gets a normal-looking sandbox where nothing real moves, and every action they try is recorded. An admin then releases the real customer or freezes the attacker, and after 30 minutes it falls back to asking for extra proof.

Feature 3, CyberMind Network: a separate small hub service where different CyberMind installs can share blocked attacker fingerprints as hashed values. A fingerprint is only shared back when at least two installs reported it, and it only adds a small warning, it never blocks on its own.

Feature 4, Adaptive stress test: a search that keeps changing an attacker's settings (number of tries, IPs, speed, browser) to find what slips past the login protection, then reports how many changes it needed and which rule was the weakest.

RAG assistant upgraded: now also produces a structured report with a CVE lookup and a fact-check of the claims.

New interface: panels open directly instead of waiting for the camera flight, the intro can be skipped, and the 3D sphere now shows real sessions as orbiting dots that change colour and movement when a session becomes suspicious, shadowed or locked.

Testing and settings: 121 automated tests, and every new feature has an on/off setting.

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How it is different from other tools

Tools like Intezer and Threat.Zone analyse files, malware and URLs. They answer the question: is this file malicious? Our project does not analyse malware at all.

The closer comparison is a behavioural fraud tool like BioCatch, which tries to tell if the person using an account is the real customer. That is the same problem we work on, but BioCatch is a commercial product used by banks, while ours is a smaller open project you can run and read yourself.

What our project does that is different:

> It connects phishing emails to login protection through the planted fake credential, so one email can lead to a blocked attacker.

> Uncertain sessions go to a sandbox instead of a simple allow or block, and we record what the attacker tries.

> Every decision shows the exact rules and reasons, not just a risk score.

> It can test itself with the attack lab and the stress test.

> Installs can warn each other without sharing any customer data.